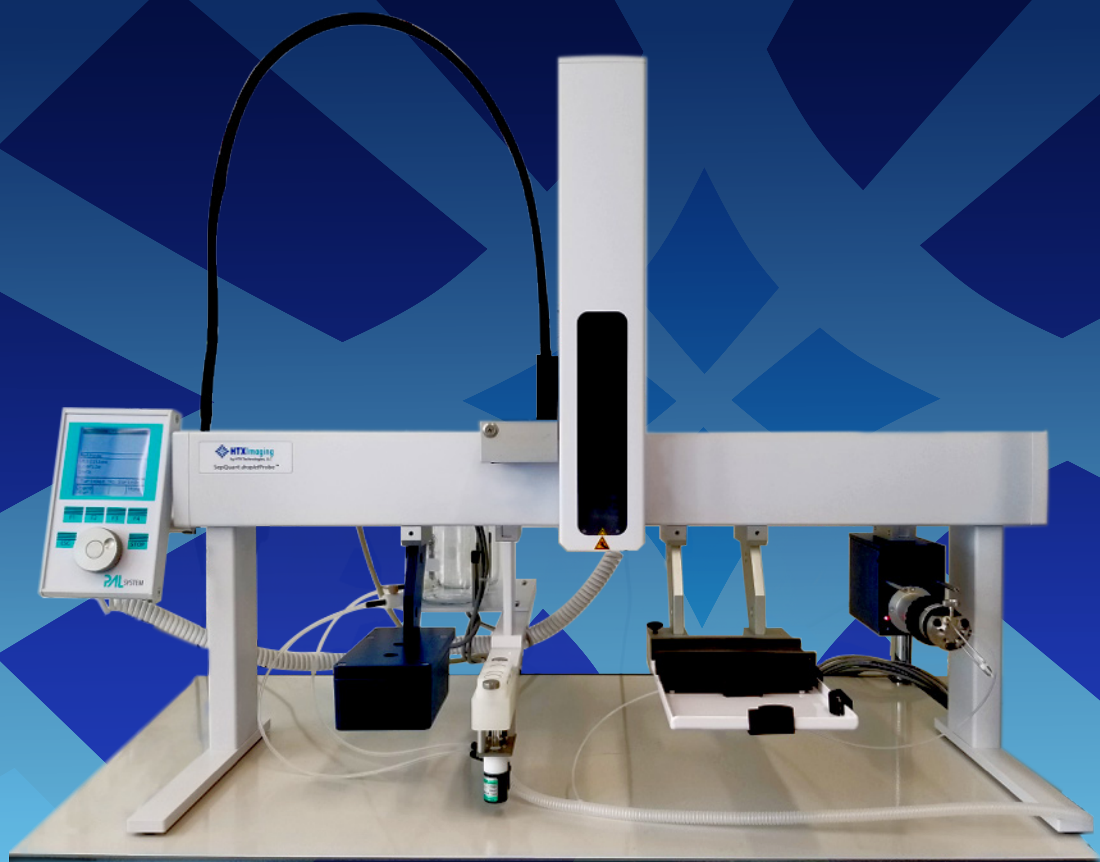
SepQuant® *droplet*Probe™

Liquid Extraction for Surface Sampling & Analysis



HTXImaging
by HTX Technologies, LLC

SepQuant® *droplet*Probe™ LESA

The SepQuant® *droplet*Probe™ Allows Liquid Surface Extraction of Analytes and In-line Separation by HPLC prior to Mass Spectrometry Analysis

The SepQuant *droplet*Probe™ is a highly stable liquid micro-junction sampling technology with patented laser sensor-to-surface controls. The *droplet*Probe™ application software integrates with the CTC PAL System offering users a reliable automation platform and integration with multiple HPLC systems and mass spectrometers.

The patented laser sensor-to-surface technology allows the precise and automatic positioning of the liquid interface. Sample height variability is no longer a limitation, and the automated calibration ensures trouble-free operation.

A custom 2-position tray allows for up to 2 MTP plates or 6 standard 1" x 3" glass slides, enabling high sample capacity. Sample vials are integrated into the hardware to allow on-the-spot micro-reactions, internal standard deposition and more.

The system is available in a 120cm wide version (HTS PAL) and 90cm (HTC PAL) to fit any size bench.

HPLC and UPLC valves are offered to allow connection to the latest chromatography systems for the highest separation performance.

Key Characteristics

- Liquid junction droplets from 0.5 to 5 µL
- Adjustable needle to surface distance (typically 100-300µm)
- Spatial resolution as low as 1 mm
- Oversampling available

Features

- Software enables:
  - Acquisition of optical images of the surface for point-and-click selection of locations to be sampled
  - Data analysis
  - Heatmap generation
  - Record keeping
- No hardware modification on the PAL System required for basic operation
- Use of a cooled stack or an open tray
- <1 min sampling time
- Method can be integrated into existing LC/MS workflow
- Manual XYZ-axis control for complex samples (e.g. fungi)
- Optional laser displacement sensor for unattended analysis of samples and sample locations of dramatically disparate height

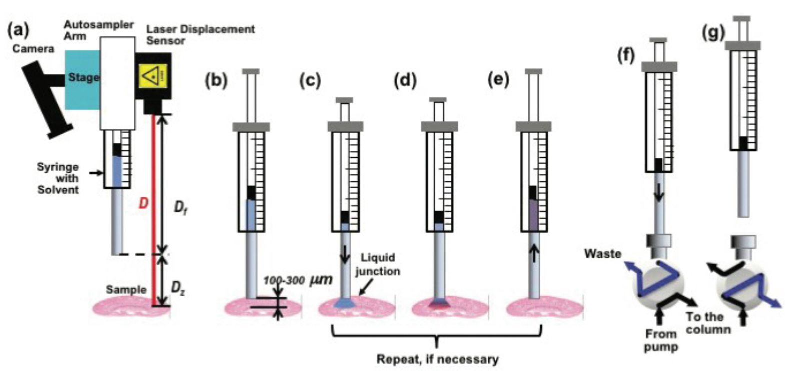


Figure 1. Schematic of the surface sampling process including (a) determining syringe-to-surface distance, (b) the positioning of the syringe needle 100-300 µm above a defined surface spot, (c) dispensing a discrete volume of extraction solvent onto a surface creating a liquid junction between the needle and the surface, (d) dissolution of the analyte in the extraction solvent, (e) the liquid is drawn back into the syringe needle after a predefined extraction time, and (f) loaded into the sample loop followed by (g) injection for a consecutive HPLC-MS/MS analysis. Steps (c)-(e) may be repeated to improve extraction efficiency.

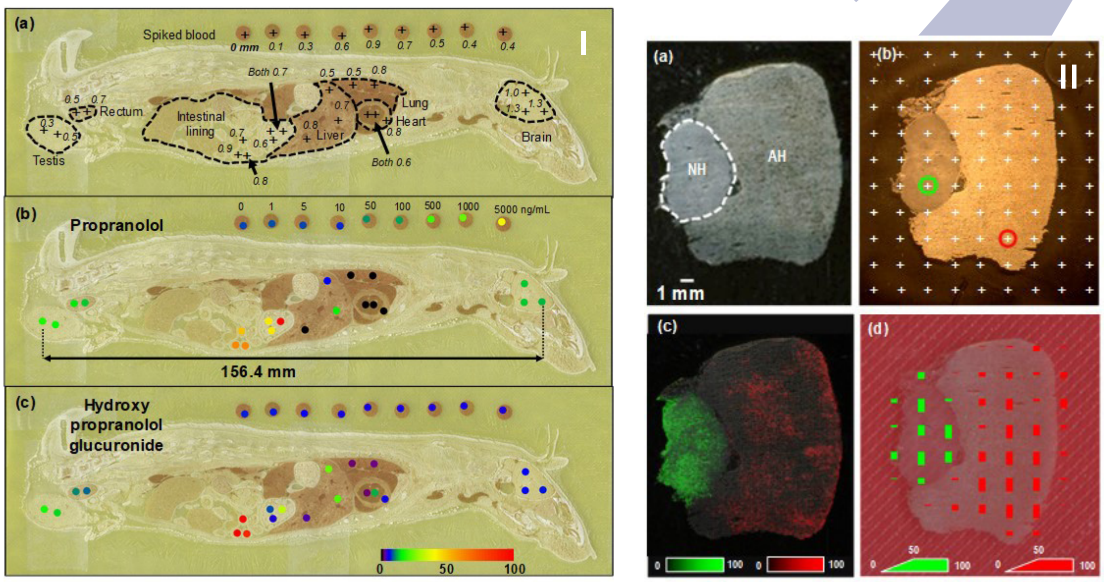


Figure 2.
(I) Heatmaps of three different analytes in whole-body rat tissue (II) (a) Optical image of a human pituitary gland thin tissue section; (b) Optical image of a sister section of sample in a. showing the sampled surface locations 2 mm apart; (c) Thin tissue section with neurohypophysis (NH) and adenohypophysis (AH) regions marked with arginine vasopressin (AVP, green) and adrenocorticotropic hormone (ACTH, red). (d) Heatmaps of AVP (green) and ACTH (red).

Applications

- Drugs and metabolites [1-4, 8-9], antibody–drug conjugate catabolites [6] and macromolecules (proteins [7] and oligonucleotides) from thin tissue sections
- Drugs from tissue biopsy [4]
- Metabolites from fungal cultures [10-12]
- Small molecules from plant tissues (e.g. capsaicin from peppers, caffeine from coffee beans)
- Drugs from over-the-counter tablets
- Small molecule contamination on drywall
- Food science and quality assurance

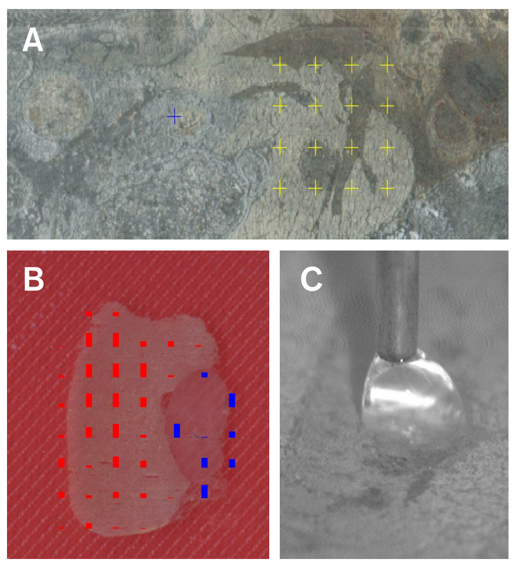


Figure 3.
(A) Easily select samples in software to (B) generate heatmaps of analytes of interest. (C) Zoomed in view of liquid microjunction between the tip of the needle and the sample taken with instrument camera for built-in quality control.

References

[1] Kertesz V, Van Berkel GJ. Anal. Chem. 82, 5917-5921 (2010).
[2] Kertesz V, Van Berkel GJ. Bioanal. 5, 819-826 (2013).
[3] Kertesz V, Paranthaman N, Moench P, Catoire A, Flarakos J, Van Berkel GJ. Bioanal. 6, 2599-2606 (2014).
[4] Kertesz V, Weiskittel TM, Van Berkel GJ. Bioanal. Chem. 407, 2117-2125 (2015).
[5] Kertesz V, Van Berkel GJ. Mass Spectrom. 28, 1553–1560 (2014).
[6] Lanshoeft C, Stutz G, Elbast W, Wolf T, Walles M, Stoeckli M, Picard F, Kretz O. Rapid Commun. Mass Spectrom. 2016, 30, 823–832
[7] Kertesz V, Calligaris D, Feldman DR, Changelian A, Laws ER, Santagata S, Agar MYR, Van Berkel GJ. Anal Bioanal Chem, 2015, 407, 5989-5998.
[8] Chen W, Wang L, Van Berkel GJ, Kertesz V, Gan J. J Chromatogr A. 2016, 1439, 137–143.
[9] Kertesz V, Weiskittel TM, Vavrek M, Freddo C, Van Berkel GJ. Rapid Commun. Mass Spectrom. 2016 Jul 30;30(14):1705-1712. doi: 10.1002/rcm.7607
[10] Sica VP, Raja HA, El-Elimat T, Kertesz V, Van Berkel GJ, Pearce CJ, Oberlies NH. J. Nat. Prod. 2015, 78, 1926–1936
[11] Sica VP, Rees ER, Tchegnon E, Bardsley RH, Raja HA, Oberlies NH. Frontiers in Microbiology, 1 April 2016, Volume 7, Article 544
[12] Paguigan ND, Raja HA, Day CS, Oberlies NH. Phytochemistry 126 (2016) 59–65

SepQuant® *droplet*Probe™ LESA

In situ Spatial Characterization of Bioactive Secondary Metabolites of *Asimini triloba* by the SepQuant® *droplet*Probe™

Efforts to identify new potential molecules for pharmacognostic research have led to many novel therapeutic discoveries in recent years. However, the field of natural products research continues to face challenges, specifically with regard to the prioritization of compounds of interest and understanding of the ecological and biological contexts of these compounds within their native organism. Using the SepQuant® *droplet*Probe™, we screened for bioactive secondary metabolites from four different organs of the plant, *Asimini triloba* (paw paw).

Chromatograms of Annonaceous acetogenins were prepared from the fruits (seed and pulp), twigs, leaves, and flowers (petal and ovary) of *A. triloba*. The in situ analysis of all of these organs by the SepQuant® *droplet*Probe™ coupled to a UPLC-HRMS/MS system enabled the rapid identification of acetogenins from complex mixtures through their characteristic MS/MS spectra.

PROTOCOL

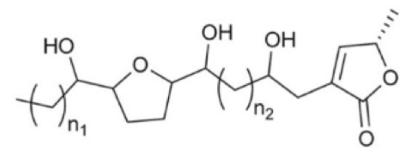
- 5 µL droplet micro-extractions were performed using MeOH-H_2O (1:1)
- A post-column lithium infusion was used in the MS analysis of *A. triloba* to promote the formation of lithiated precursor and product ions.

INSTRUMENTATION

Sample Prep:	SepQuant® *droplet*Probe™
Analysis:	Acquity ultra-performance liquid chromatography (UPLC) system coupled to an electrospray QExactive MS

A

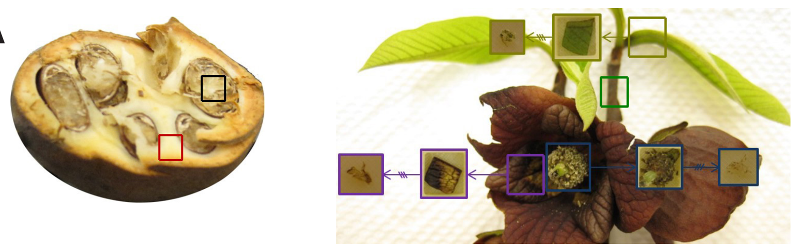
B



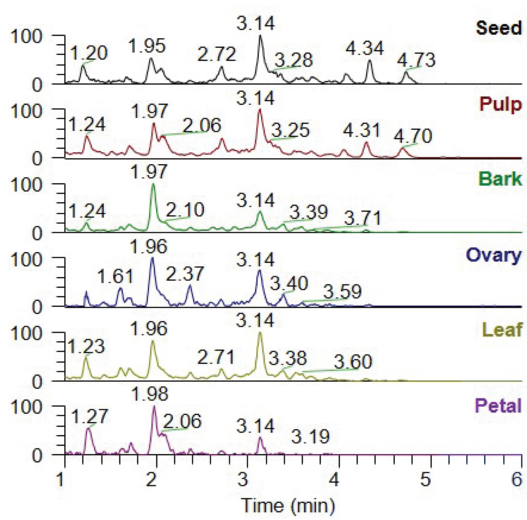
n_1 = 9-13
n_2 = 4-12

(**A**) Photograph of *Asimina triloba* and magnification of the fruit.
(**B**) General backbone for the mono-tetrahydrofuran structural class of Annonaceous acetogenins.

A



B



(**A**) Locations of paw paw where the *droplet*Probe™ directly sampled seed (black), pulp (red), and twig (green) and the portions that were cross-sectioned: ovary (blue), leaf (yellow), and petal (purple).

(**B**) The mass defect filtered chromatograms around annonacin; 603.4807 +/- 100 Da with a mass defect of +/- 25 mDa.

SepQuant® *droplet*Probe™ is available worldwide exclusively from HTX Technologies, LLC. To request further information contact:

Alain Creissen
Imaging Product Manager, HTX Technologies
acreissen@htximaging.com